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Quality Assessment Statistic Evaluation of X-Ray Fluorescence via NIST and IAEA Standard Reference Materials

O. M. H. Ahmed[1,2*], Farouk I. Habbani[3], A. M. Mustafa[2], E. M. A. Mohamed[4], A. M. Salih[1], Ftihia Seedig[3]

[1]Chemistry and Nuclear Physics Institute, Sudan Atomic Energy Commission, Khartoum, Sudan
[2]College of Nuclear Science and Technology, Harbin Engineering University, Harbin China
[3]Department of Physics, Faculty of Science, University of Khartoum, Khartoum, Sudan
[4]Council of Atomic Energy, Sudan Academy of Sciences, Khartoum, Sudan
Email: *osama@hrbeu.edu.cn, *osama.alagra@yahoo.com

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Abstract

The aim of this study is a quality assessment of X-ray fluorescence laboratory located at the University of Khartoum. The X-ray fluorescence spectrometer system consists, a set of three ^{109}Cd sources of an initial nominal activity of 10 μCi, and Si(Li) detector Energy Dispersive XRF(EDXRF) systems. It is important to carry out this work because it has an effective contribution for a wide range of research and services. The assessment was carried out by measuring 8 NIST-2709a (soil) and 13 IAEA-155 (milk powder) standard reference material samples for repeatability examinations to test the measurement precision. The total combined standards uncertainty values for XRF lab were estimated by an error from repeatability measurements adding 2.6% for error propagation related to the method. For accuracy assessment, three standard statistic approaches were applied, *i.e.* the Bias %, zeta-score, and E_n-number. The bias of all elements for both standard materials was found to be within a deviation range from −28% to 7.8%. The results of all elements for both the zeta-score test and E_n-number have satisfactory results except Th (Thorium) and Zr (Zirconium) which consider as questionable results for NIST SRM 2709a and unsatisfactory results for E_n-number.

Keywords

XRF, QC, Reference Material, Statistic Evaluation

1. Introduction

X-ray fluorescence analysis (XRF) is a powerful physical technique, which is a